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COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

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Quiz #1

September 11, 2025

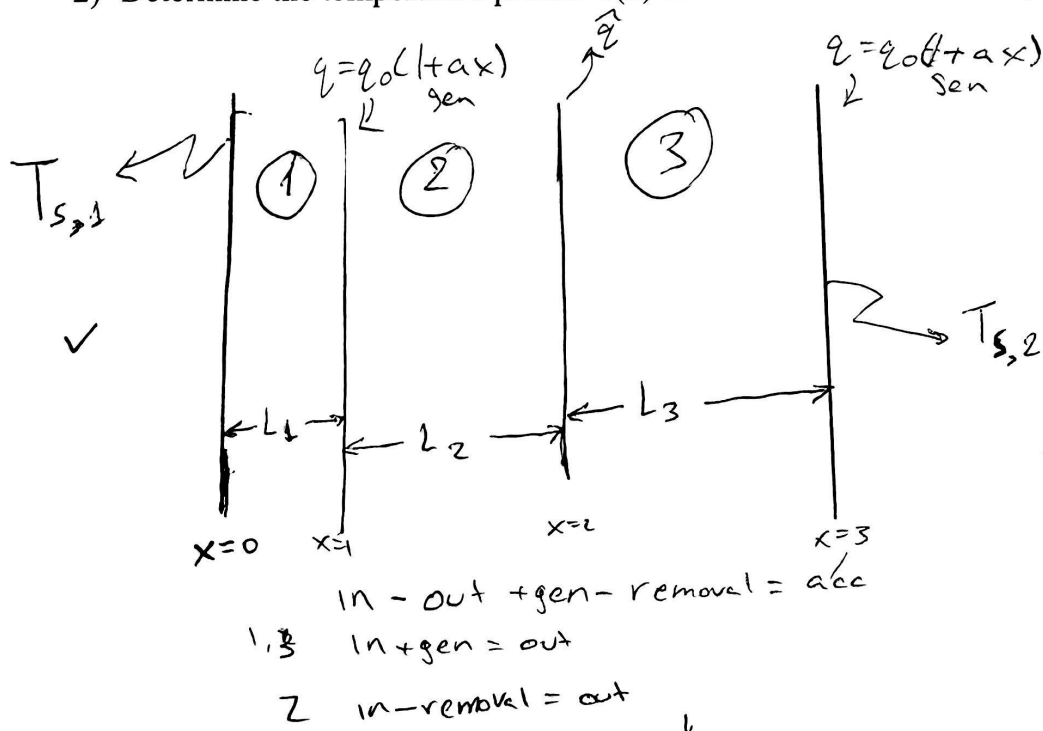
Open Notes and Book

Duration: 40 minutes

Consider three sequential one-dimensional plane walls of thickness L_1 , L_2 and L_3 , respectively; heat is being transferred through this set of plane walls since the surface temperatures of the walls are maintained at $T_{s,1}$ and $T_{s,2}$ as shown in the sketch. The first and third ones experience $\dot{q} = \dot{q}_0(1 + a \cdot x)$ volumetric heat generation; the middle one involves uniform volumetric heat removal $\hat{q}$. All properties are constant.

1) Write all design/balance equations and corresponding boundary conditions

2) Determine the temperature profile $T(x)$ and heat flux distribution $q''(x)$.



$$1) x=0 \quad T=T_{s,1}$$

$$\checkmark x=L_1 \quad T = k_1 \frac{dT(x)}{dx} = k_2 \frac{dT(x)}{dx}$$

$$x=L_2 \quad T = k_2 \frac{dT_2(x)}{dx} = k_3 \frac{dT_3(x)}{dx}$$

$$x=L_3 \quad T=T_{s,2}$$

$$x=L_3 \quad T_3=T_{s,2}$$

$$x=L_1 \quad T_1|_{x=L_1} = T_2|_{x=L_2}$$

$$\int \frac{q_0 + q_0 a x}{k_1} dx$$

$$\frac{q_0}{k_1} \int (1 + ax) dx$$

$$\frac{q_0}{k_1} \left(x + \frac{1}{2} ax^2 \right) + A$$

$$2) \quad k_2 \frac{d^2 T_2(x)}{dx^2} = \overset{\text{removed}}{q_0} \rightarrow \frac{d^2 T_2(x)}{dx^2} = \frac{q_0}{k_2} \rightarrow \frac{dT_2(x)}{dx} = \frac{q_0}{k_2} x + A \rightarrow T_2(x) = \frac{1}{2} \frac{q_0}{k_2} x^2 + \underline{A} x + \underline{B}$$

$$k_1 \frac{d^2 T_1(x)}{dx^2} = q_0(1+ax) \rightarrow \frac{d^2 T_1(x)}{dx^2} = - \frac{q_0(1+ax)}{k_1}$$

$$\left\{ \frac{dT_1(x)}{dx} = - \frac{q_0}{k_1} \left(x + \frac{1}{2} ax^2 \right) + E \right.$$

$$T_1(x) = - \frac{q_0}{k_1} \left(\frac{1}{2} x^2 + \frac{1}{6} ax^3 \right) + Ex + F$$

$$\text{Similar integral for } \frac{dT_3(x)}{k_3 dx} = -q_0(1+ax)$$

$$T_1(x=0) \rightarrow T = \boxed{T_{s,1} = B} \quad \text{BC1}$$

$$T(x=L_1) \rightarrow \boxed{k_1 \left(- \frac{q_0}{k_1} \left(L_1 + \frac{1}{2} a L_1^2 \right) + E \right) = k_2 \left(\frac{q_0}{k_2} L_1 + A \right)} \quad \text{BC2}$$

$$T_3(x=L_3) = - \frac{q_0}{k_3} \left(\frac{1}{2} L_3^2 + \frac{1}{6} a L_3^3 \right) + Gx + J$$

$$\boxed{ - \frac{q_0}{k_3} \left(\frac{1}{2} L_3^2 + \frac{1}{6} a L_3^3 \right) + G(L_3) + J = T_{s,2} } \quad \text{BC4}$$

$$\boxed{ T_2(x=L_2) \quad k_2 \left(\frac{q_0}{k_2} L_2 + A \right) = k_3 \left(- \frac{q_0}{k_3} \left(L_3 + \frac{1}{2} a L_3^2 \right) + G \right) }$$

$$k_3 \frac{d^2 T_3(x)}{dx^2} = -q_0(1+ax) \rightarrow \frac{d^2 T_3(x)}{dx^2} = - \frac{q_0(1+ax)}{k_3} \rightarrow \frac{dT_3(x)}{dx} = - \frac{q_0}{k_3} \left(x + \frac{1}{2} ax^2 \right) + G$$

$$T_3(x) = - \frac{q_0}{k_3} \left(\frac{1}{2} x^2 + \frac{1}{6} ax^3 \right) + Gx + J$$

$$\boxed{ T_{s,1} = - \frac{q_0}{k_1} \left(\frac{1}{2} L_1^2 + \frac{1}{6} a L_1^3 \right) + E L_1 + F = \frac{1}{2} \frac{q_0}{k_2} L_1^2 + A L_1 + B }$$